



Mini Simulation Project (MSP) Task Report Document

Group Name:	AMM	Date Submitted:	Nov 28, 2025
Project Title:	AUTOMATIC INK REFILLER		
Leader:	Ephraim Flores	Task:	☐ Docu ☐ HW ☐ SW
Members:	Alexa Gomez	Task:	☐ Docu ☐ HW ☐ SW
	Riley Sy	Task:	☐ Docu ☐ HW ☐ SW

I. Team/Member Evaluation

This Automatic Marker Refiller project stands as a result of the essential role of teamwork and collective effort in engineering design and implementation. The successful development of this system, from its initial hardware architecture based on the 8086 MPU and 8255 PPIs, through the intricate process of creating the control software in Assembly language, to the final testing and integration of the ADC, pumps, and LCD was only made possible through the dedicated efforts and specialized contributions of all three members of this group.

Flores, Ephraim Joshua M.

| Team Leader

Ephraim served as the team leader and played a crucial role in shaping the overall direction of the project. He drafted the initial setup for the PPIs and 8259 PIC, establishing the foundation for how the system handled interrupts and I/O operations. He also designed the early system architecture, specifically deciding that the **ON/OFF switch** would be the primary interrupt source, while the **START** function would be handled through standard PPI input, avoiding interrupt conflicts at a time when interrupt prioritization had not yet been introduced in class.

The project itself is heavily based on his earlier Arduino prototype for an automated ink refiller, which he adapted into a full 8086 microprocessor-based system. He engineered the **limit-switch-triggered motor system**, enabling four markers to be refilled simultaneously through concatenated slot flags. Ephraim also refined the **ink-level detection design**, choosing **1V** as the lower threshold and mapping the ADC's **51/255** step reading as the minimum operational limit.

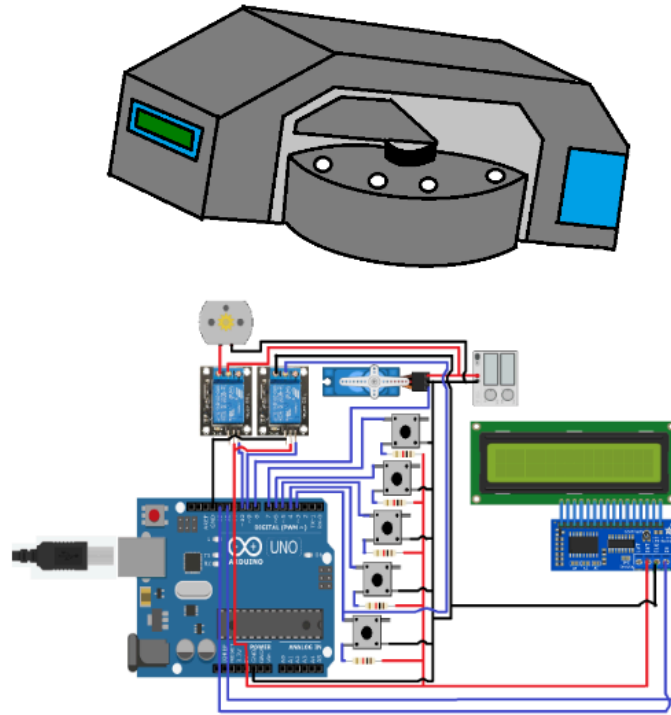


Fig 1. Ephraim's original Arduino-based prototype concept that inspired the 8086 microprocessor implementation.

Another significant contribution was his continuous loop structure for monitoring marker slot switches and driving indicator LEDs, ensuring real-time feedback. He also conducted extensive simulations involving the **555 timer**, **8086**, and **8253** for servo-based PWM, and although each attempt proved unworkable, these experiments directly influenced the team's final decision to adopt **fixed-position pumps**, improving stability and durability.

Gomez, Alexa Janine

| Member

Alexa contributed significantly to the conceptual and documentation aspects of the project. She refined and polished the system flowchart, ensuring that the logic sequence and operational steps were clear and aligned with the final implementation. She also assisted in developing the LCD interface by helping test display sequences and verifying message outputs.

In addition, Alexa served as the primary organizer for hardware images and simulation files in Proteus, maintaining a repository that the team used for reporting and verification. She prepared Canva-based presentations and frequently asked technical questions to strengthen the team's explanatory materials for the project defense.

Alexa also contributed to the team's research on PWM generation by simulating both the **555 timer** and pulse outputs directly from the **8086**. While neither approach succeeded in driving a servo motor reliably, her experiments played a key role in concluding that a servo-driven mechanism was unsuitable. This guided the team toward adopting **fixed-position pump actuators**, improving reliability and extending the lifespan of the device by avoiding components prone to mechanical wear and electrical sensitivity.

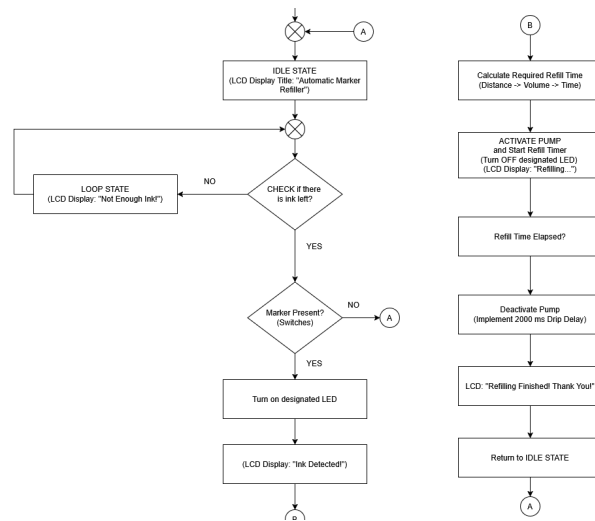


Fig 2. Alexa's initial draft flowchart outlining the system's refill logic and decision flow

Sy, Riley Aaron O.

| Member

Riley played a central role in building the complete hardware foundation of the project. She designed and assembled the full 8086 microprocessor environment, integrating the 8284 clock generator, 8288 bus controller, address latches, data transceiver, and LS138 address decoder responsible for mapping all peripherals into their designated port ranges. Her hardware layout ensured clean bus timing, proper address demultiplexing, and stable PPI access, forming the electrical backbone required for the system to function reliably.

She also configured and wired both 8255 PPIs, assigning their Port A, B, and C lines to the LCD interface, ADC control pins, motor drivers, and limit-switch inputs. By checking direction bits, handshake expectations, and mode configurations, she ensured that each command word issued in software matched the actual electrical setup. This prevented I/O mapping conflicts and allowed seamless data and control flow between the processor, ADC, motors, and indicator devices.

In addition to hardware integration, Riley developed the full LCD subsystem and its command routines, documenting the wiring, timing constraints, and initialization sequence. She prepared small diagnostic programs, verified ADC-to-PPI signal flow, and coordinated subsystem testing to ensure everything behaved consistently before full software integration. Her work provided the stable and well-verified hardware environment that allowed the system's higher-level logic to operate smoothly.

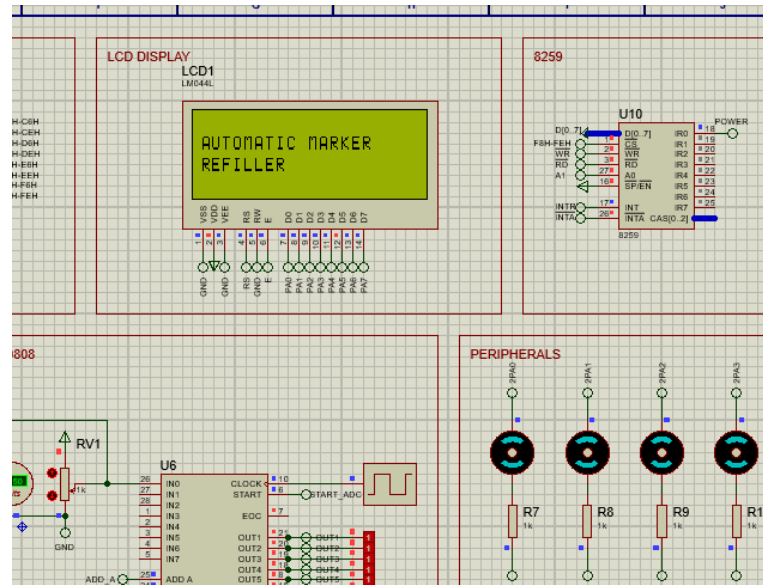


Fig 3. Riley's integrated hardware layout featuring the LCD and core 8086 peripherals.